

Pavithran Gnanasekaran

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EDUCATION

State University of New York at Buffalo

Aug 2024 - Dec 2025

- M.S. in Computer Science – AI & ML

Visvesvaraya Technological University

Jun 2024 - May 2025

- B.E. in Electronics & Communication

Coursework: DS & Algorithms, Deep Learning, Computer Vision, Database Systems, Distributed Systems, Operating Systems

EXPERIENCE & INTERSHIPS

Alvora AI

Dover, Delaware

Software Engineer

Aug 2025 - Present

- Built RAG chatbot using LLMs with retrieval pipeline, validated on 200 test queries, improving answer relevance by 40%
- Engineered a mentor-matching engine using cosine similarity + Elasticsearch, reducing matching time from 8s to 2.8s
- Optimized prompt pipelines & containerized Django services with Docker/K8s, cutting env setup time to under 2 mins.
- Developed Django frontend to collect user data, validate user and store in DB, securing 250+ user & Mentor data.

University at Buffalo

Buffalo, New York

Research Assistant

May 2025 - Present

- Built 16 node multi agent pipeline on HPC cloud for Explainable Medical AI achieved 90% report generation accuracy
- Deployed realtime RAG application via RestAPI & Hugging Face using Docker/Kubernetes cutting latency by 30%
- Designed GraphRAG reasoning engine with LangChain/LangGraph & Neo4j improving retrieval precision by 40%

UB-Office of Academic Affairs

Buffalo, New York

Software Engineer- AI & Infra Platforms

Jan 2025 - May 2025

- Engineered N8N-based agentic workflow to query DB, enrich leads and automate 2700+ LinkedIn and email outreach.
- Designed & implemented data pipelines using Apache Spark, Kafka & Airflow, reduced worktime from 6 hr to 90 min
- Refactored 12 slow PostgreSQL queries reducing avg. execution time from 4.2s to 0.8s improving accuracy by 10%
- Monitored three microservices on Kubernetes with Docker & AWS EKS ensuring 99% uptime under peak load.

RESEARCH PAPER - IEEE 7th International Conference on Cognitive Machine Intelligence ([Link](#))

Self-Supervised MRI Representation Learning for Cross-Domain Brain Disorder Classification

Pittsburgh, Pennsylvania

First-Author

IEEE CogMI 2025

PROJECTS

TaskFlow Pro – Full-Stack Task Management Application ([Link](#))

Jun 2025 - Aug 2025

- Designed a Full-stack React–FastAPI–MongoDB system with JWT auth, supporting 100% task lifecycle via REST APIs.
- Integrated Email notification alerts + group AI chat assistant, reducing task failure rate from 12% to 2% via retry logic.
- Dockerized entire full-stack app and reduced setup time to ~1 min and deployed via Render + Vercel CI/CD pipeline.

Real-Time SSD Failure Detection using LSTM Autoencoders ([Link](#))

May 2025 – Jun 2025

- Designed an LSTM autoencoder for SSD anomaly detection on 3.1M+ SMART records, achieving ~98% failure recall.
- Modeled anomalies via reconstruction error thresholds and detected 1578/1580 drive failures before crash events.
- Reduced 90+ SMART features to 18 principal components via PCA, cutting training time & improving efficiency by 40%

LLM-Based Text Summarization NLP Models with BART ([Link](#))

Mar 2025 - May 2025

- Built BART+RAG summarizer handling 4K words, generating concise 100–200 words outputs with contextual retrieval.
- Trained with PyTorch AMP & Hugging Face, reducing GPU memory by 35% and accelerating convergence time by 1/3.
- Deployed FastAPI backend on Railway app with Docker, Nginx and MongoDB with 200ms latency for 100+ requests.

SKILLS

Languages/Databases: Python, C/C++, Java, Ruby, JavaScript, C#, Typescript, MySQL, MongoDB, Firebase, PostgreSQL

Web/ App Development: HTML/CSS, Flask, Django, FastAPI, Angular, React, NodeJs, TailwindCSS, ExpressJs

AI/Cloud Technologies: PyTorch, TensorFlow, LangChain, LangGraph, RAG, Docker, Kubernetes, AWS, GCP, Kafka, Terraform

Other: Software Development Life cycle, Stanford (CS 230), GitHub, Tableau, Git, Linux, Jira, Bash.